

Augmentation of Tabular Data for Boosting Accuracy

Digital Health M – Group Hands-on Project P3

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Abstract

The report consists of a walkthrough of our group hands-on project. We chose track P3: Augmentation of Tabular Data for Boosting Accuracy, it is about tabular data augmentation on a given dataset. The challenge is an increased difficulty in data augmentation on tabular data. The objective is to classify patients as inpatient or outpatient by analyzing CBC features and then understand if data augmentation can be beneficial or not in improving predictive performance.

We compared three approaches: a baseline Random Forest classifier, SMOTE augmentation, and Noise Injection augmentation. The results show that the baseline model achieved the best overall accuracy and weighted F1-score, while SMOTE slightly improved macro F1-score and macro recall. Noise Injection produced results close to the baseline but did not provide a significant improvement. The report also discusses the clinical plausibility of synthetic tabular data and the limitations of augmentation in medical datasets.

1. Project Objective

The main objective of the project is determining if data augmentation can be beneficial for CBC-based ML classification, while keeping clinical plausibility.

We trained a model to classify patients as either **inpatient** (**in**) or **outpatient** (**out**). We compared three different approaches:

1. a baseline model trained without augmentation;
2. a model trained with SMOTE augmentation;
3. a model trained with Noise Injection augmentation.

2. Dataset Description

The dataset used in this project is `dataset_project3.csv`. It contains **3000 patient records** and **11 columns**.

The target variable is `SOURCE`, which indicates the patient source and has two possible classes:

- **in**: inpatient;
- **out**: outpatient.

The input features include CBC-related numerical variables and demographic variables.

Table 1: Dataset features

Feature	Type	Description
HAEMATOCRIT	Numerical	Hematocrit value
HAEMOGLOBINS	Numerical	Hemoglobin concentration
ERYTHROCYTE	Numerical	Red blood cell count
LEUCOCYTE	Numerical	White blood cell count
THROMBOCYTE	Numerical	Platelet count
MCH	Numerical	Mean Corpuscular Hemoglobin
MCHC	Numerical	Mean Corpuscular Hemoglobin Concentration
MCV	Numerical	Mean Corpuscular Volume
AGE	Numerical	Patient age
SEX	Categorical	Patient sex
SOURCE	Categorical	Target class

The dataset does not contain missing values.

The class distribution is moderately imbalanced, as shown in Table 2.

Table 2: Class distribution in the full dataset

Class	Count	Percentage
out	1874	62.47%
in	1126	37.53%

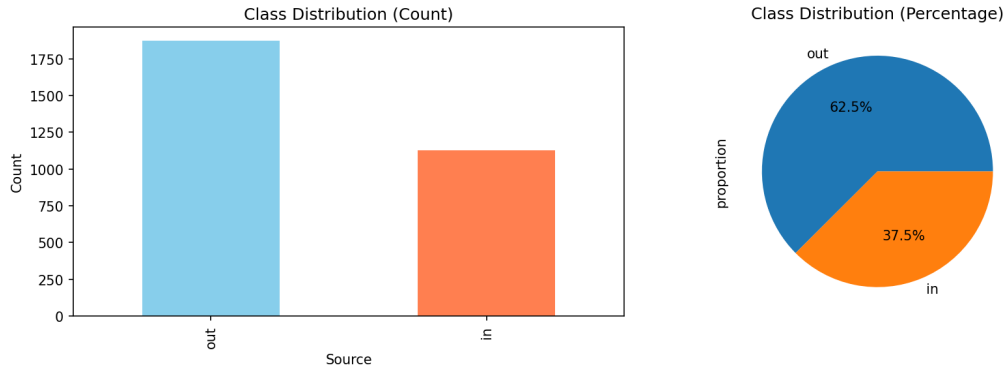


Figure 1: Class distribution.

This imbalance is important because a model may achieve good overall accuracy while still performing worse on the minority class, which is the inpatient class.

3. Experimental Setup

The dataset was split into training and test sets using an 80/20 split. Stratification was applied in order to preserve the original class distribution in both sets.

Table 3: Train-test split

Split	Samples
Training set	2400
Test set	600

The class distribution in the training and test sets is shown in Table 4.

Table 4: Class distribution after splitting

Class	Training samples	Test samples
in	901	225
out	1499	375

A **Random Forest classifier** was used for all three experiments. The same random seed and evaluation protocol were used across all models to make the comparison reproducible.

Categorical variables were encoded using **OneHotEncoder**, while numerical variables were standardized where needed inside the preprocessing pipeline.

All augmentation techniques were applied **only to the training set**. The test set was never modified.

For the baseline model, we tried the Logistic Regression algorithm as well, but it performed slightly worse than the Random Forest Classifier. Hence, we decided to refer to the Random Forest as our baseline model.

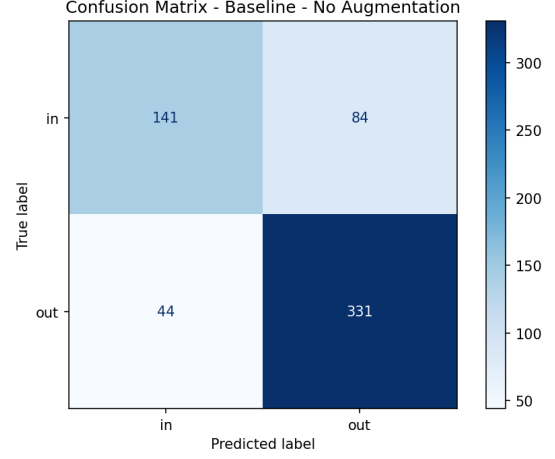
4. Baseline Model Performance

The baseline model was trained on the original training set without any data augmentation.

The baseline Random Forest achieved the performance shown in Table 5 on the untouched test set.

Table 5: Baseline model performance

Metric	Score
Accuracy	0.7867
Precision weighted	0.7843
Recall weighted	0.7867
F1 weighted	0.7817
Precision macro	0.7799
Recall macro	0.7547
F1 macro	0.7629

**Figure 2:** Baseline model confusion matrix.

The baseline model obtained the highest overall accuracy and weighted F1-score among the three tested approaches. However, the macro metrics show that the model did not perform equally well on both classes, which is expected because the dataset is moderately imbalanced.

5. SMOTE Augmentation

SMOTE stands for **S**ynthetic **M**inority **O**versampling **T**echnique and is an interpolation-based augmentation method.

Instead of simply duplicating minority-class samples, SMOTE creates new synthetic examples by interpolating between existing minority-class observations. In our project, we used SMOTE to generate additional synthetic samples for the minority class in the training set.

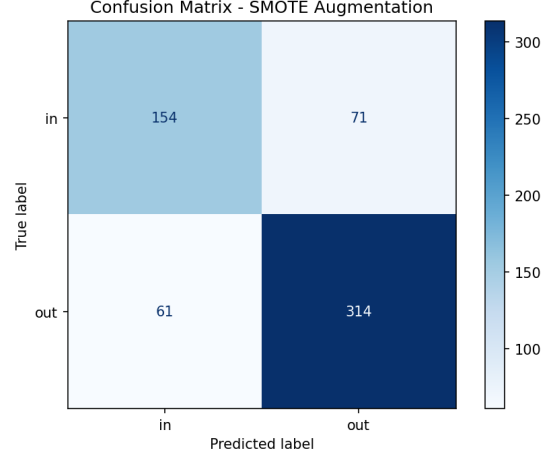
The goal of SMOTE was to make the classifier less biased toward the majority class and improve the detection of inpatient cases.

A limitation of this SMOTE implementation is that SMOTE was applied after preprocessing, including One-Hot encoding of the categorical variable **SEX**. SMOTE generates synthetic samples through numerical interpolation, this means it may produce intermediate values in the One-Hot encoded columns, such as values between 0 and 1. These values are not directly interpretable as real patient categories. Hence, some SMOTE-generated samples may not correspond to clinically realistic patient profiles.

After training the Random Forest classifier on the SMOTE-augmented training set, the model obtained the results shown in Table 6.

Table 6: SMOTE model performance

Metric	Score
Accuracy	0.7800
Precision weighted	0.7783
Recall weighted	0.7800
F1 weighted	0.7789
Precision macro	0.7659
Recall macro	0.7609
F1 macro	0.7632

**Figure 3:** SMOTE confusion matrix.

When compared with the baseline, SMOTE slightly decreased accuracy and weighted F1-score. However, it slightly improved macro recall and macro F1-score. This suggests that SMOTE helped the model treat the two classes more evenly, even though it did not improve the overall predictive performance.

6. Noise Injection Augmentation

Noise Injection is a perturbation-based augmentation technique. It creates synthetic samples by adding small random noise to numerical features.

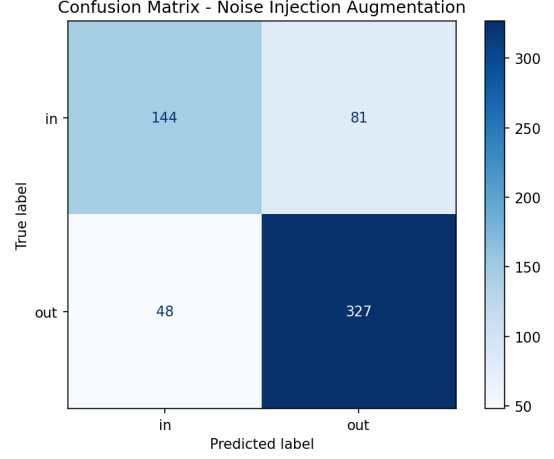
In this project, noise was added only to numerical features in the training set that are not in relationship with each other (MCH, MCHC and MCV has been excluded). The noise level was kept low in order to avoid generating clinically unrealistic CBC values. The categorical feature **SEX** was not modified: it has been OneHotEncoded, so if noise had been applied there would have been uncosistent values (ex. since $M=0$ and $F=1$, if noise is added we would have values like $M=0.3$, $F=0.7$. **SEX** data would lose their true meaning).

The purpose of Noise Injection was to make the model more robust by exposing it to slightly varied versions of the original training samples.

The Noise Injection model obtained the results shown in Table 7.

Table 7: Noise Injection model performance

Metric	Score
Accuracy	0.7850
Precision weighted	0.7822
Recall weighted	0.7850
F1 weighted	0.7810
Precision macro	0.7757
Recall macro	0.7560
F1 macro	0.7629

**Figure 4:** Noise Injection confusion matrix.

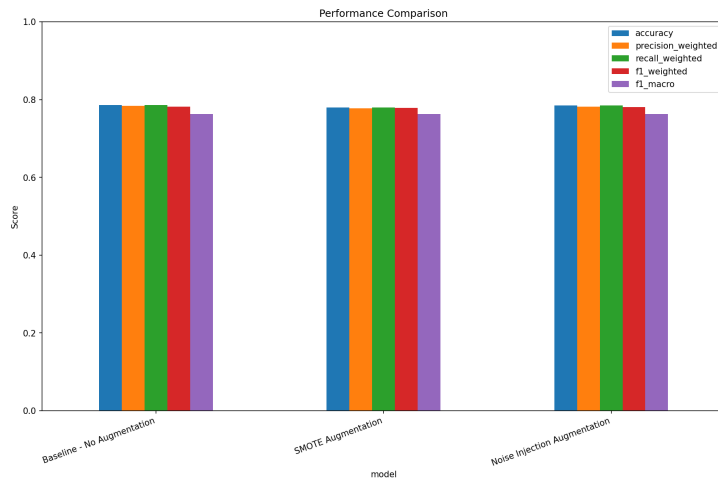
Noise Injection did not clearly improve performance. Its results were close to the baseline but slightly worse in most metrics, except for macro recall which performed better than baseline and F1-macro that is equal to the value of the baseline. This suggests that the added perturbations did not provide enough useful new information to improve the classifier.

7. Performance Comparison

Table 8 summarizes the results of the three approaches.

Table 8: Performance comparison between baseline and augmented models

Model	Accuracy	Precision W.	Recall W.	F1 W.	Precision M.	Recall M.	F1 M.
Baseline – No Augmentation	0.7867	0.7843	0.7867	0.7817	0.7799	0.7547	0.7629
SMOTE Augmentation	0.7800	0.7783	0.7800	0.7789	0.7659	0.7609	0.7632
Noise Injection Augmentation	0.7850	0.7822	0.7850	0.7810	0.7757	0.7560	0.7629

**Figure 5:** Performance Comparison.

Overall, the baseline model achieved the best accuracy and weighted F1-score. SMOTE achieved the best macro recall and macro F1-score, although the difference was very small. Noise Injection produced results close to the baseline but did not significantly improve the model.

8. Discussion

8.1. Which method performed best, and why?

The best model in terms of overall performance was the **baseline Random Forest without augmentation**. It achieved the highest accuracy and weighted F1-score.

However, SMOTE produced the highest macro F1-score and slightly improved macro recall. This is important because macro metrics give equal importance to both classes. Therefore, although SMOTE did not improve overall accuracy, it helped the classifier become slightly more balanced between inpatient and outpatient cases.

This results show that the “best” method depends on the metric considered. If the goal is to maximize global accuracy, the baseline model performed best. If the goal is to improve balance between classes, SMOTE was the most interesting augmentation method.

Noise Injection did not provide a clear advantage. One possible explanation is that the perturbation level was intentionally small to preserve clinical plausibility. As a result, the generated samples may have been too similar to the original data to significantly improve learning.

8.2. Did augmentation introduce unrealistic data?

Augmentation may introduce unrealistic data, especially in medical tabular datasets.

CBC variables are physiologically related. For example, MCH (Mean Corpuscular Hemoglobin) and MCHC (Mean Corpuscular Hemoglobin Concentration) relate to HAEMOGLOBINS and HAEMATOCRIT, MCV (Mean Corpuscular Volume) should relate to HAEMATOCRIT and ERYTHROCYTE. A synthetic sample may contain values that are individually within a reasonable range but unrealistic when considered together.

For Noise Injection, this risk was reduced by adding only small perturbations and avoiding changes to categorical variables. However, this does not fully guarantee clinical plausibility, because the relationships between different blood parameters may still be altered.

For SMOTE, synthetic samples are generated by interpolation. This may produce plausible intermediate values, but it may also create artificial patient profiles that do not correspond to real clinical cases. A limitation of our implementation is that SMOTE was applied after preprocessing, including OneHot encoding of the categorical variable **SEX**. This may create synthetic encoded values that are useful for the classifier but not directly interpretable as real patient profiles.

A more clinically faithful alternative could be to use SMOTENC, which is designed to handle

both numerical and categorical variables.

8.3. Is augmentation always beneficial in medical datasets?

No, augmentation is not always beneficial in medical datasets.

In this experiment, augmentation did not clearly improve the overall performance of the classifier. The baseline model achieved the best accuracy and weighted indexes, while SMOTE only slightly improved macro F1-score and macro recall. Noise Injection did not improve the model.

This shows that synthetic data generation must be used carefully in healthcare. Medical data are complex, and feature relationships often have clinical meaning. If augmentation creates unrealistic samples, the model may learn patterns that are not clinically valid.

Therefore, augmentation should not be evaluated only through numerical metrics. It should also be assessed in terms of clinical plausibility, safety, and trustworthiness.

9. Conclusion

This project compared a baseline Random Forest classifier with two tabular data augmentation techniques: SMOTE and Noise Injection.

The baseline model achieved the best overall performance, with the highest accuracy and weighted indexes. SMOTE did not improve global performance, but it slightly improved macro F1-score and macro recall, suggesting a small benefit in balancing the performance between inpatient and outpatient cases. Noise Injection produced results close to the baseline but did not provide a clear improvement.

The main conclusion is that tabular data augmentation in healthcare is not automatically beneficial. While it can help address class imbalance, it may also introduce synthetic patient profiles that are difficult to validate clinically.

For this reason, augmentation in medical datasets should always be applied only to the training set, evaluated with multiple metrics, and interpreted carefully in relation to the clinical meaning of the features.